

Q1.

```
import java.util.*;
class Q1 {
    public static void main(String[] args){

        Scanner input =new Scanner(System.in);

        System.out.print("Input First Number : ");
        int f_num = input.nextInt();

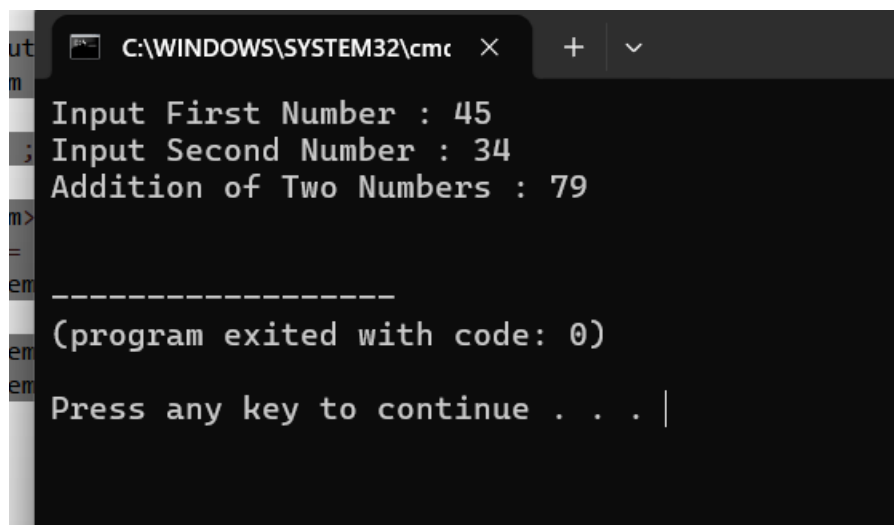
        System.out.print("Input Second Number : ");
        int s_num = input.nextInt();

        int add ;

        if (f_num>s_num){
            add = f_num + s_num ;
            System.out.println("Addition of Two Numbers : "+add);
        }else{
            System.out.println("First Number : "+f_num);
            System.out.println("Second Number : "+s_num);
        }

    }
}
```

Output>>



```
C:\WINDOWS\SYSTEM32\cmd.exe
Input First Number : 45
Input Second Number : 34
Addition of Two Numbers : 79
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q2.

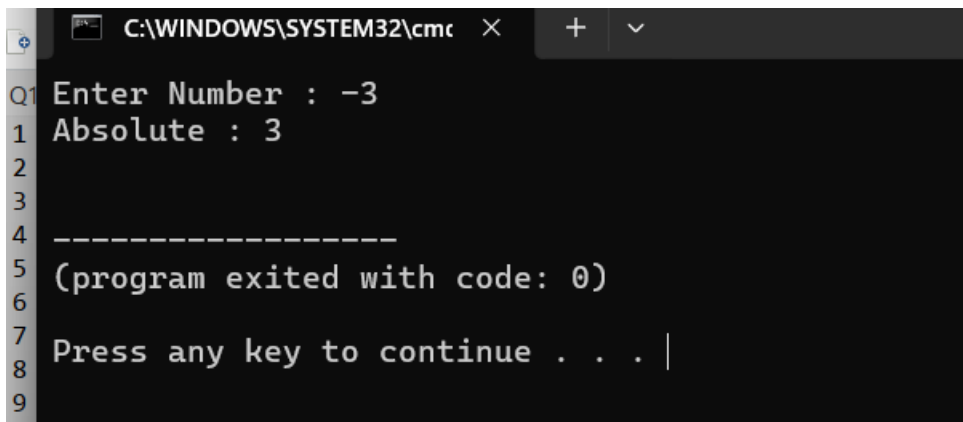
```
import java.util.*;
class Q2 {
    public static void main(String[] args){

        Scanner input=new Scanner(System.in);

        System.out.print("Enter Number : ");
        int number =input.nextInt();
        int abs;
        if (number<0){
            abs= -number;
            System.out.println("Absolute : "+abs);
        }else{
            abs = number ;
            System.out.println("Absolute : "+abs);
        }

    }
}
```

Output>>



```
Q1 Enter Number : -3
1 Absolute : 3
2
3
4 -----
5 (program exited with code: 0)
6
7 Press any key to continue . . . |
8
9
```

Q3.

```
import java.util.*;
class Q3 {
    public static void main(String[] args){

        Scanner input =new Scanner(System.in);

        System.out.print("Enter Chemistry Marks      : ");
        double chem_marks = input.nextInt();

        System.out.print("Enter Physics Marks      : ");
        double physi_marks = input.nextInt();

        System.out.print("Enter Combined Maths Marks : ");
        double com_marks = input.nextInt();

        double total = chem_marks + physi_marks + com_marks ;
        double average = total/3;
        System.out.println("Total : "+total);
        System.out.println("Average : "+average);
        if(average>75){
            System.out.println("Pass");
        }else{
            System.out.println("Fail");
        }
    }
}
```

Output>>

```
Enter Chemistry Marks      : 78
Enter Physics Marks      : 69
Enter Combined Maths Marks : 90
Total : 237.0
Average : 79.0
Pass
```

****Pass***

```
Enter Chemistry Marks      : 67
Enter Physics Marks      : 56
Enter Combined Maths Marks : 45
Total : 168.0
Average : 56.0
Fail
```

****Fail***

Q4.

```
import java.util.*;
class Q4 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

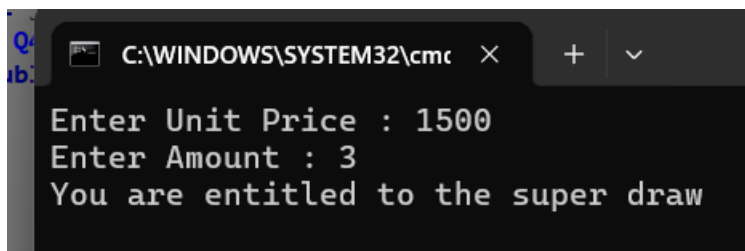
        System.out.print("Enter Unit Price : ");
        double uni_price = input.nextDouble();

        System.out.print("Enter Amount : ");
        int amount = input.nextInt();

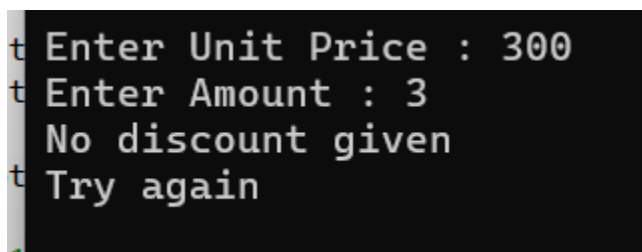
        double total = uni_price*amount;

        if(total>1500){
            System.out.println("You are entitled to the super draw");
        }else{
            System.out.println("Try again ");
        }
    }
}
```

Output>>



```
C:\WINDOWS\SYSTEM32\cmd
Enter Unit Price : 1500
Enter Amount : 3
You are entitled to the super draw
```



```
Enter Unit Price : 300
Enter Amount : 3
No discount given
Try again
```

Q5.

```
import java.util.*;
class Q5 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

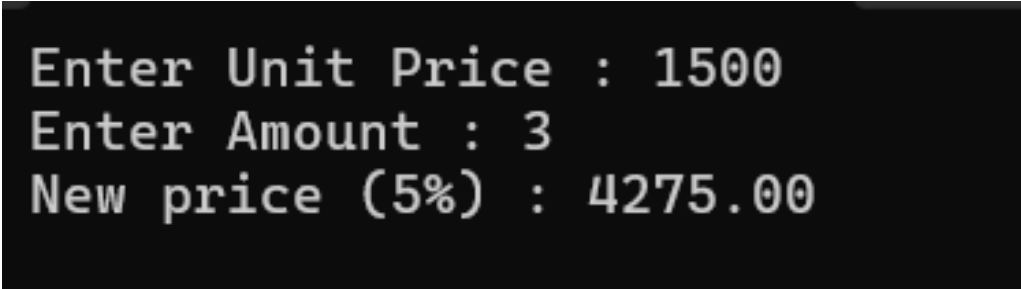
        System.out.print("Enter Unit Price : ");
        double unit_price = input.nextDouble();

        System.out.print("Enter Amount : ");
        int amount = input.nextInt();

        double total = unit_price*amount;
        double discount;
        double ne_total;
        if(total>500){
            discount = total*0.05;
            ne_total = total-discount;
            System.out.printf("New price (5%%) : %.2f \n",ne_total);

        }else{
            System.out.println("No discount give");
        }
    }
}
```

Output>>

A screenshot of a terminal window with a black background and white text. It shows the output of the Java program: 'Enter Unit Price : 1500', 'Enter Amount : 3', and 'New price (5%) : 4275.00'.

```
Enter Unit Price : 1500
Enter Amount : 3
New price (5%) : 4275.00
```

Q6.

```
import java.util.*;
class Q6 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a Random Year : ");
        int y_num = input.nextInt();

        if(y_num%4 == 0 && y_num % 100 !=0 || (y_num % 400 ==0 )){
            System.out.println("It is a leap Year ");
        }else{
            System.out.println("It is not a leap Year ");
        }
    }
}
```

Output>>

```
Enter a Random Year : 2008
It is a leap Year
```

****leap year***

```
Enter a Random Year : 2007
It is not a leap Year
```

not a leap Year

Q7.

```
import java.util.*;
class Q7 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Input number 01 : ");
        int num_1 =input.nextInt();

        System.out.print("Input number 02 : ");
        int num_2 =input.nextInt();

        System.out.print("Input number 03 : ");
        int num_3 =input.nextInt();

        int max_num =num_1;

        if(num_2>max_num){
            max_num = num_2;
        }
        if(num_3>max_num){

            max_num = num_3;
        }
        System.out.println("The Maxumum Number is : "+max_num);
    }
}
```

Output>>

```
Input number 01 : 67
Input number 02 : 45
Input number 03 : 98
The Maxumum Number is : 98

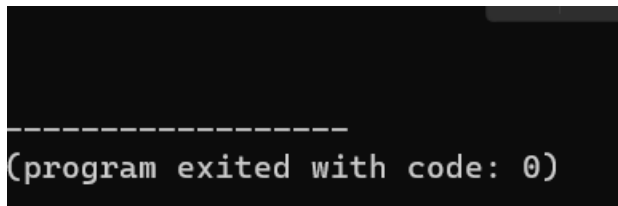
-----
(program exited with code: 0)
```

Q8.

```
class Q8 {  
    public static void main(String[] args) {  
        int x = 10; //line3  
        if(x==10){}  
    }  
}
```

Answer>> C. *if(x==10){}*

Output>>



```
-----  
(program exited with code: 0)
```

****it will not execute any output.***

Q9.

```
class Q9 {  
    public static void main(String[] args) {  
        int x = 10;  
        boolean b=true;  
        if(){} // A,B,C,E,F are correct  
    }  
}
```

Answers>>

A -if(b){}

B -if(b=flase){}

C-if (b==false){}

E-if((b==false)==false){}

F-if(b=(false==true)){}

Q10.

```
import java.util.*;
class Q10 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter an integer number : ");
        int num = input.nextInt();

        if(num%2 == 0){
            System.out.println("The number is an even number");

        }else{
            System.out.println("The number is an odd number");
        }
    }
}
```

Output>>

```
Enter an integer number : 25
The number is odd number
```

****odd number***

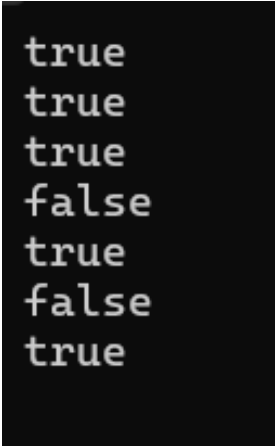
```
Enter an integer number : 20
The number is even number
```

****even number***

Q11.

```
class Q11 {  
    public static void main(String[] args) {  
        boolean b=true;  
        System.out.println(b);  
        System.out.println(b=true);  
        System.out.println(b==true);  
        System.out.println(b!=true);  
        System.out.println(b=true==true);  
        System.out.println((b=true)==false);  
        System.out.println(b=(true!=false));  
    }  
}
```

Output>>



```
true  
true  
true  
false  
true  
false  
true
```

Q12.

```
class Q12 {  
    public static void main(String[] args) {  
        int x=10;  
        System.out.println(x=9);  
        System.out.println(x==9);  
        System.out.println(x=9!=10); //this line is incorrect  
        System.out.println((x=9)==10);  
        System.out.println((x=9)<=10);  
    }  
}
```

```
javac "Q12.java" (in directory: D:\Rush\ICD124\day05\Assignment05)  
Q12.java:6: error: incompatible types: boolean cannot be converted to int  
        System.out.println(x=9!=10);  
                        ^  
1 error  
Compilation failed.
```

Output>>

- **Without line 6**

```
9  
true  
false  
true
```

Q13.

```
class Q13{
    public static void main(String[] args){
        int x=20,y=60;
        boolean bool;
        System.out.println(x=10);
        System.out.println(bool=true);

        System.out.println(x=10>0); // this is incorrect
        System.out.println((x=10)>0);
        System.out.println(bool=(x=10)>0);
        System.out.println(bool=x+y>100);
    }
}
```

```
javac "Q13.java" (in directory: D:\Rush\ICD124\day05\Assignment05)
Q13.java:7: error: incompatible types: boolean cannot be converted to int
        System.out.println(x=10>0); // this is incorrect
                        ^
1 error
Compilation failed.
```

Output without line 7

```
5
5 10
5 true
5 true
5 true
5 false
```

Q14.

```
class Q14{
    public static void main(String[] args){
        byte b = 10;
        short s = 100;
        int x = 125;
        long l = 15000;
        float f = 1.5f;
        double d = 21.231;
        char c = 'c';
        boolean bool = 10>9;

        System.out.println(b+s+x+f+d+c+bool); //Line 1 output
        System.out.println(b+s+x+f+d+c+bool); //Line 2
        System.out.println(b+s+x+f+d+c+bool); //Line 3
        System.out.println(b+s+x+f+d+c+bool); //Line 4
        System.out.println(b+s+x+f+d+c+bool); //Line 5
    }
}
```

Line4,5 Error

```
javac Q14.java (in directory: D:\Kush\ICD124\day05\Assignments)
Q14.java:14: error: bad operand types for binary operator '+'
        System.out.println(b+s+x+f+d+c+bool+ ""); //Error
                                ^
first type: double
second type: boolean
Q14.java:15: error: bad operand types for binary operator '+'
        System.out.println(b+s+x+f+d+c+bool+ ""); //Error
                                ^
first type: boolean
second type: byte
2 errors
Compilation failed.
```

Without line 4,5

```
2351.521.231ctrue
101001251.521.231ctrue
356.731true
```

Q15.

```
class Q15{
    public static void main(String[] args){
        int i = 1,j = 2,k =3,m = 2 ;
        System.out.println( i ==1); //Line 1
        System.out.println( j ==3); //Line 2
        System.out.println( ( i >=1) && ( j <4) ); //Line 3
        System.out.println( ( m <=99) & ( k < m ) ); //Line 4
        System.out.println( ( j >= i ) || ( k == m ) ); //Line 5
        System.out.println( ( k + m < j )||(3- j>= k)); //Line 6
        System.out.println( !( k > m ) ); //Line 7
    }
}
```

Output>>

```
true
false
true
false
true
false
false
```

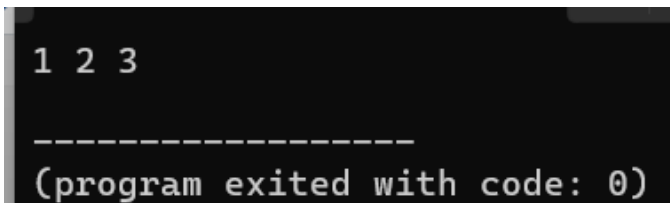
Q16.

(A.) *int x=1;*

```
class Q16{
    public static void main(String args[]){
        int x=1;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:



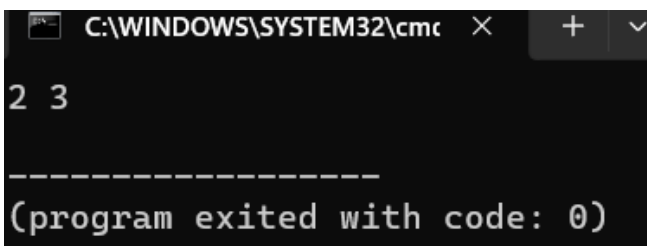
```
1 2 3
-----
(program exited with code: 0)
```

(B.) *int x=2;*

```
class Q16{
    public static void main(String args[]){
        int x=2;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:



```
2 3
-----
(program exited with code: 0)
```

(C.) *int x =3;*

```
class Q16{
    public static void main(String args[]){
        int x=3;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:

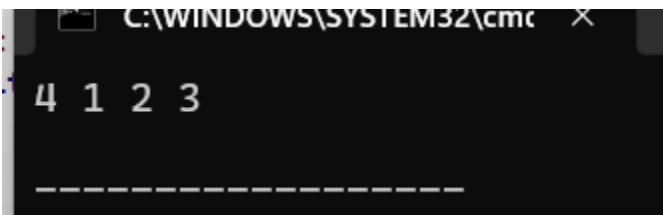
A screenshot of a terminal window with a black background. The number '3' is printed in white. Below it is a horizontal dashed line. The rest of the terminal content is obscured by a dark overlay.

(D.) *int x =4;*

```
class Q16{
    public static void main(String args[]){
        int x=4;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:

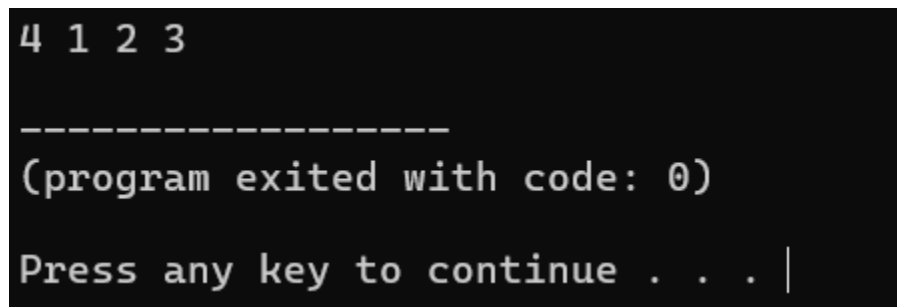
A screenshot of a terminal window with a black background. The sequence '4 1 2 3' is printed in white. Below it is a horizontal dashed line. The rest of the terminal content is obscured by a dark overlay.

(E.) *int x = 0;*

```
class Q16{
    public static void main(String args[]){
        int x=4;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:

A screenshot of a terminal window showing the output of a Java program. The first line is "4 1 2 3". The second line is a separator consisting of 15 dashes. The third line is "(program exited with code: 0)". The fourth line is "Press any key to continue . . . |" with a vertical cursor bar at the end.

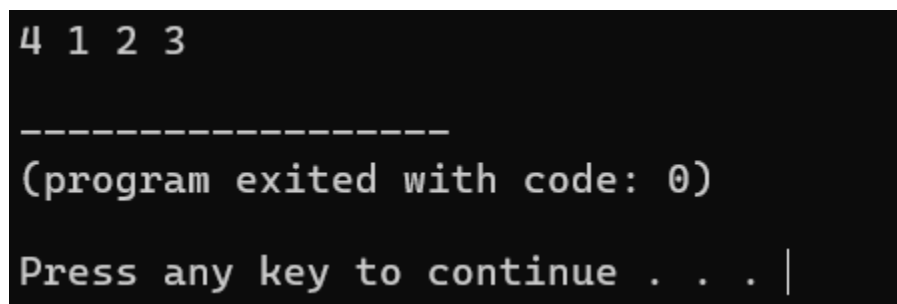
```
4 1 2 3
-----
(program exited with code: 0)
Press any key to continue . . . |
```

(F.) *int x = 5;*

```
class Q16{
    public static void main(String args[]){
        int x=4;
        switch(x){
            default: System.out.print("4 ");
            case 1 : System.out.print("1 ");
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");

        }
    }
}
```

Output:

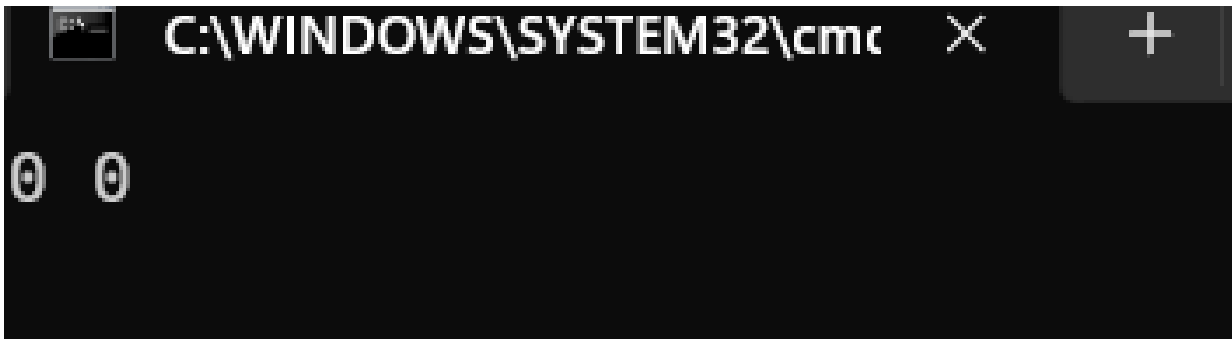
A screenshot of a terminal window showing the output of a Java program. The first line is "4 1 2 3". The second line is a separator consisting of 15 dashes. The third line is "(program exited with code: 0)". The fourth line is "Press any key to continue . . . |" with a vertical cursor bar at the end.

```
4 1 2 3
-----
(program exited with code: 0)
Press any key to continue . . . |
```

Q17.

```
class Q17{  
    public static void main(String args[]){  
        int a=-5;  
        int b=-2;  
        a%=b;  
        a/=b;  
        b=a>0?a;  
        System.out.println(a+" "+b);  
    }  
}
```

Output:



Q18.

```
import java.util.*;
class Q18{
    public static void main(String args[]){
        int x=100;
        System.out.println(x); //Line 1
        {
            int y=200;
            {
                int z=300;
                System.out.println(x); //Line 2
                System.out.println(y); //Line 3
                System.out.println(z); //Line 4
            }
            System.out.println(x); //Line 5
            System.out.println(y); //Line 6
            //System.out.println(z); //Line 7
        }
        System.out.println(x); //Line 8
        //System.out.println(y); //Line 9
        //System.out.println(z); //Line 10
    }
}
```

**Answers are: A: line 1, B: line2 ,C:line3, D:line 4, E:line5 ,
F: line 6, H: line8**

Q19.

(A.)

```
class Q19{
    public static void main(String args[]){
        int x=1;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

Output>>

A screenshot of a Java program's output. It shows a black rectangular window with the number '1' printed in white text on the left side. The window has a standard title bar with a close button.

(B.)

```
class Q19{
    public static void main(String args[]){
        int x=2;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

Output>>

A screenshot of a Java program's output. It shows a black rectangular window with the numbers '2 3 1' printed in white text on the left side. The window has a standard title bar with a close button and a file path 'C:\WINDOWS\SYSTEM32' visible in the background.

(C)

```
class Q19{
    public static void main(String args[]){
        int x=3;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

Output>>

A terminal window with a black background and white text. The text "3 1" is displayed on the first line.

(D)

```
class Q19{
    public static void main(String args[]){
        int x=4;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

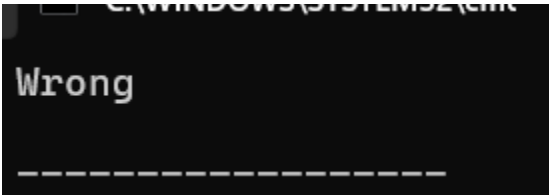
Output>>

A terminal window with a black background and white text. The text "Wrong" is displayed on the first line, followed by a dashed line on the second line.

(E)

```
class Q19{
    public static void main(String args[]){
        int x=0;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

Output>>

A screenshot of a terminal window with a black background. The word "Wrong" is printed in a light gray, monospaced font. Below it is a horizontal dashed line of the same color. The top of the window shows a portion of a file path: "C:\WINDOWS\SYSTEM32\cmd".

(F)

```
class Q19{
    public static void main(String args[]){
        int x=5;
        switch(x){
            default : System.out.print("Wrong ");break;
            case 2 : System.out.print("2 ");
            case 3 : System.out.print("3 ");
            case 1 : System.out.print("1 ");break;
        }
    }
}
```

Output>>

A screenshot of a terminal window with a black background. The word "Wrong" is printed in a light gray, monospaced font. Below it is a horizontal dashed line of the same color.

Q20.

```
import java.util.*;
class Q20 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter an integer : ");
        double num = input.nextInt();

        if(num<0){
            System.out.println(" Negative integer ");

        }else if(num>0){
            System.out.println(" Positive integer ");
        }else{
            System.out.println(" Zero ");
        }
    }
}
```

Output:

```
Enter an integer : -3
Negative integer
```

```
Enter an integer : 4
Positive integer
```

```
Enter an integer : 0
Zero
```

Q21.

```
import java.util.*;
class Q21 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

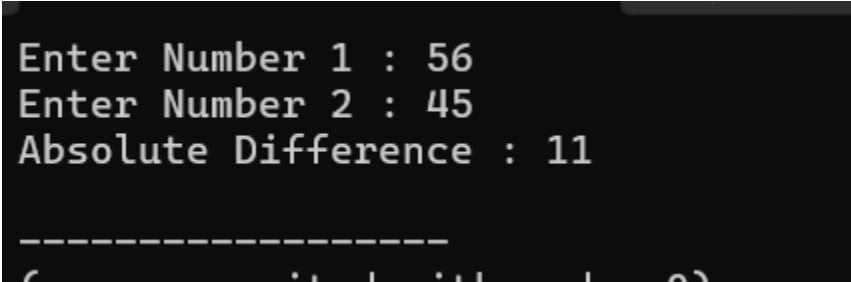
        System.out.print("Enter Number 1 : ");
        int num_1 = input.nextInt();

        System.out.print("Enter Number 2 : ");
        int num_2 = input.nextInt();

        int absolute_difference ;
        if (num_1>num_2){
            absolute_difference = num_1 - num_2 ;

        }else{
            absolute_difference = num_2 - num_1 ;
        }
        System.out.print("Absolute Difference : "+absolute_difference );
    }
}
```

Output:

A screenshot of a terminal window with a black background and white text. It shows the output of the Java program: 'Enter Number 1 : 56', 'Enter Number 2 : 45', and 'Absolute Difference : 11'. Below the output, there is a dashed line and a partially visible line of text.

```
Enter Number 1 : 56
Enter Number 2 : 45
Absolute Difference : 11
-----
6. ... .. 2)
```

Q22.

```
import java.util.*;
class Q22 {
    public static void main(String[] args){
        //line 4
        switch(x){
            case 'A':System.out.println("65");break;
            case 'B':System.out.println("66");break;
            case 'C':System.out.println("67");break;
            default :System.out.println("wrong");
        }
    }
}
```

Answer: A,B,C,D,E

Q23.

```
import java.util.*;
class Q23 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter First Number : ");
        int f_num = input.nextInt();

        System.out.print("Enter Second Number : ");
        int s_num = input.nextInt();

        if(f_num<s_num){
            System.out.println("The first number is less than the second number");
        }else if (f_num>s_num){
            System.out.println("The First number is greater than the second number");
        }else{
            System.out.println("Both are equal");
        }
    }
}
```

Outputs:

```
Enter First Number : 25
Enter Second Number : 39
The first number is less than the second number
```

```
Enter First Number : 39
Enter Second Number : 25
The First number is greater than the second number
```

```
Enter First Number : 25
Enter Second Number : 25
Both are equal
```

Q24.

```
import java.util.*;
class Q24 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter First Integer Number : ");
        int f_num = input.nextInt();

        System.out.print("Enter Second Integer Number : ");
        int s_num = input.nextInt();

        System.out.print("Enter Third Integer Number : ");
        int third_num = input.nextInt();

        int digit1 = f_num %10 ;
        int digit2 = s_num %10 ;
        int digit3 = third_num %10 ;

        if(digit1 == digit2 || digit1 == digit3 || digit2 == digit3 ){
            System.out.println("True");
        }else{
            System.out.println("False");
        }
    }
}
```

Outputs:

```
Enter First Integer Number : 36
Enter Second Integer Number : 16
Enter Third Integer Number : 14
True
```

```
s Enter First Integer Number : 13
t Enter Second Integer Number : 26
s Enter Third Integer Number : 38
t False
```

Q25.

```
Q11.java ✕ Q12.java ✕ Q13.java ✕ Q14.java ✕ Q15.java ✕ Q16.java ✕ Q17.java ✕ Q18.java ✕ Q19.java
import java.util.*;
class Q25 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter First Integer Number : ");
        int f_num = input.nextInt();

        System.out.print("Enter Second Integer Number : ");
        int s_num = input.nextInt();

        System.out.print("Enter Third Integer Number : ");
        int third_num = input.nextInt();

        // Calculate absolute differences
        int abs1 = (s_num > third_num) ? (s_num - third_num) : (third_num - s_num);
        int abs2 = (f_num > third_num) ? (f_num - third_num) : (third_num - f_num);
        int abs3 = (f_num > s_num) ? (f_num - s_num) : (s_num - f_num);

        // Check if one is greater than the difference of others
        if (f_num > abs1) {
            System.out.println("First number Greater than subtraction of others :True ");
        }else{
            System.out.println("First number greater than subtraction of others :False ");
        }
        if(s_num > abs2){
            System.out.println("Second number greater than subtraction of others :True ");
        }else{
            System.out.println("Second number greater than subtraction of others :False ");
        }
        if(third_num > abs3){
            System.out.println("Thrid number greater than subtraction of others :True ");
        }else{
            System.out.println("Thrid number greater than subtraction of others :False ");
        }
    }
}
```

Output:

```
Enter First Integer Number : 15
Enter Second Integer Number : 65
Enter Third Integer Number : 35
First number greater than subtraction of others :False
Second number greater than subtraction of others :True
Thrid number greater than subtraction of others :False
```

Q26.

```
import java.util.*;
class Q26 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the selling price : ");
        double sell_price =input.nextDouble();

        System.out.print("Enter cost of product : ");
        double cost=input.nextDouble();

        if(sell_price > cost){
            System.out.println("Profit");
        }else if (sell_price< cost){
            System.out.println("loss");
        }else{
            System.out.println("No Profit No loss");
        }
    }
}
```

Output:

```
Enter the selling price : 20000
Enter cost of product : 12000
Profit
```

```
Enter the selling price : 20000
Enter the selling price : 2000
Enter cost of product : 2000
No Profit No loss
```

Q27.

```
import java.util.*;
class Q27 {
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter First integer number : ");
        double num_1 = input.nextInt();

        System.out.print("Enter Secod integer number : ");
        double num_2 = input.nextInt();

        System.out.print("Enter Third integer number : ");
        double num_3 = input.nextInt();

        if(num_1>num_2 && num_2>num_3){
            System.out.print("Decreasing");
        }else if(num_3>num_2 && num_2>num_1){
            System.out.print("Increasing");
        }else{
            System.out.print("Neither increasing nor decreasing order");
        }
    }
}
```

Output:

```
Enter First integer number : 34
Enter Secod integer number : 45
Enter Third integer number : 67
Increasing
```

```
Enter First integer number : 45
Enter Secod integer number : 35
Enter Third integer number : 25
Decreasing
```

```
Enter First integer number : 24
Enter Secod integer number : 56
Enter Third integer number : 23
Neither increasing nor decreasing order
```

Q28.

```
import java.util.Scanner;
class Q28 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your age    : ");
        int age = input.nextInt();

        System.out.print("Enter your weight : ");
        double weight = input.nextDouble();

        if (age > 18 && weight >= 50) {
            System.out.println("You are eligible to donate blood.");
        } else {
            System.out.println("You are not eligible to donate blood.");
        }
    }
}
```

Outputs:

```
Enter your age    : 20
Enter your weight : 67
You are eligible to donate blood.
```

```
Enter your age    : 13
Enter your weight : 35
You are not eligible to donate blood.
```

Q29.

```
import java.util.*;
class Q29 {
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter First Number : ");
        double f_num = input.nextDouble();

        System.out.print("Enter Second Number : ");
        double s_num = input.nextDouble();

        if ((f_num > 0 && s_num > 0) || (f_num < 0 && s_num < 0)) {
            System.out.println("true");
        } else {
            System.out.println("false");
        }
    }
}
```

Output:

```
Enter First Number : 4
Enter Second Number : 5
true
```

```
Enter First Number : -2
Enter Second Number : 4
false
```

Q30.

```
Q14.java Q15.java Q16.java Q17.java Q18.java Q19.java
import java.util.*;
class Q30 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = input.next().charAt(0);

        if (ch >= 'a' && ch <= 'z') {
            System.out.println("It is a lowercase letter.");
        }
        else if (ch >= 'A' && ch <= 'Z') {
            System.out.println("It is an uppercase letter.");
        }
        else {
            System.out.println("It is not a letter.");
        }
    }
}
```

Output:

```
Enter a character: A
It is an uppercase letter.
```

```
Enter a character: a
It is a lowercase letter.
```

Q31.

```
import java.util.*;
class Q31 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = input.nextInt();

        if (num % 7 == 0 || num % 10 == 7) {
            System.out.println("It is a Buzz Number.");
        } else {
            System.out.println("It is not a Buzz Number.");
        }
    }
}
```

Output:

```
Enter a number: 7
It is a Buzz Number.
```

```
Enter a number: 5
It is not a Buzz Number.
```

Q32.

```
import java.util.Scanner;
class Q32 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter total number of classes held: ");
        double Classes = input.nextDouble();

        System.out.print("Enter number of classes attended: ");
        double attended_Classes = input.nextDouble();

        double attendance_Percentage = (attended_Classes / Classes) * 100;
        System.out.println("Attendance Percentage: " + attendance_Percentage + "%");

        if (attendance_Percentage >= 70) {
            System.out.println("You are allowed to sit for the exam.");
        } else {
            System.out.print("Do you have a medical cause? (Y/N): ");
            char medical = input.next().charAt(0);

            if (medical == 'Y' || medical == 'y') {
                System.out.println("You are allowed to sit for the exam due to medical causes.");
            } else {
                System.out.println("You are not allowed to sit for the exam.");
            }
        }
    }
}
```

Output:

```
Enter total number of classes held: 50
Enter number of classes attended: 20
Attendance Percentage: 40.0%
Do you have a medical cause? (Y/N): y
You are allowed to sit for the exam due to medical causes.
```

```
Enter total number of classes held: 50
Enter number of classes attended: 40
Attendance Percentage: 80.0%
You are allowed to sit for the exam.
```

Q33.

```
import java.util.*;
class Q33 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("is employee permanent (Y/N) : ");
        char per_emp = input.next().charAt(0);

        double salary = 0;
        double y_service = 0;
        double bonus_Percentage = 0;

        if (per_emp == 'Y' || per_emp == 'y') {
            System.out.print("Enter your salary: ");
            salary = input.nextDouble();

            System.out.print("Enter your years of service : ");
            y_service = input.nextDouble();

            if (y_service < 5) {
                bonus_Percentage = 0.10;
            } else if (y_service < 10) {
                bonus_Percentage = 0.15;
            } else {
                bonus_Percentage = 0.25;
            }

            double bonus_Amount = salary * bonus_Percentage;
            double t_salary = salary + bonus_Amount;

            System.out.println("Bonus amount: " + bonus_Amount);
            System.out.println("Total salary with bonus: " + t_salary);
        } else {
            System.out.println("Not eligible for bonus (Employee is not permanent).");
        }
    }
}
```

Outputs:

```
is employee permanent (Y/N) : y
Enter your salary: 6000
Enter your years of service : 5
Bonus amount: 900.0
Total salary with bonus: 6900.0
```

```
is employee permanent (Y/N) : y
Enter your salary: 60000
Enter your years of service : 10
Bonus amount: 15000.0
Total salary with bonus: 75000.0
```

```
is employee permanent (Y/N) : y
Enter your salary: 45000
Enter your years of service : 20
Bonus amount: 11250.0
Total salary with bonus: 56250.0
```

```
is employee permanent (Y/N) : n
Not eligible for bonus (Employee is not permanent).
```

Q34.

```
import java.util.*;
class Q34 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter number of books: ");
        int quantity = input.nextInt();

        double subtotal = quantity * 100.0;
        double discount = 0;

        if (subtotal > 5000) {
            discount = subtotal * 0.10;
        }

        double total = subtotal - discount;

        System.out.println("Subtotal : " + subtotal);
        if (subtotal > 5000) {
            System.out.println("Discount : " + discount);
        } else {
            System.out.println("Discount :      -");
        }
        System.out.println("TOTAL      : " + total);
    }
}
```

Outputs:

```
Enter number of books: 25
Subtotal : 2500.0
Discount :      -
TOTAL      : 2500.0
```

```
Enter number of books: 60
Subtotal : 6000.0
Discount : 600.0
TOTAL      : 5400.0
```

Q35.

```
import java.util.*;;
class Q35 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter temperature: ");
        int temp = input.nextInt();

        if (temp >= 80) {
            System.out.println("Swimming");
        } else if (temp >= 60 && temp < 80) {
            System.out.println("Tennis");
        } else if (temp >= 40 && temp < 60) {
            System.out.println("Golf");
        } else {
            System.out.println("Skiing");
        }
    }
}
```

Outputs:

```
Enter temperature: 90
Swimming
```

```
Enter temperature: 45
Golf
```

```
Enter temperature: 60
Tennis
```

```
Enter temperature: 10
Skiing
```

Q36.

```
import java.util.*;
class Q36 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = input.next().charAt(0);

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
            System.out.println("It is a vowel.");
        } else {
            System.out.println("It is a consonant.");
        }
    }
}
```

Outputs:

```
s Enter a character: a
a It is a vowel.
```

```
Enter a character: f
It is a consonant.
```

Q37.

```
import java.util.*;
class Q37 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter current annual salary: ");
        double salary = input.nextDouble();

        System.out.print("Enter performance rating (1=excellent, 2=good, 3=poor): ");
        int rating = input.nextInt();

        double raisePercentage = 0;

        if (rating == 1) {
            raisePercentage = 0.06;
        } else if (rating == 2) {
            raisePercentage = 0.04;
        } else if (rating == 3) {
            raisePercentage = 0.015;
        }

        double raiseAmount = salary * raisePercentage;
        double newSalary = salary + raiseAmount;

        System.out.println("Raise amount: " + raiseAmount);
        System.out.println("New salary: " + newSalary);
    }
}
```

Outputs:

```
Enter current annual salary: 60000
Enter performance rating (1=excellent, 2=good, 3=poor): 1
Raise amount: 3600.0
New salary: 63600.0
```

```
Enter current annual salary: 40000
Enter performance rating (1=excellent, 2=good, 3=poor): 2
Raise amount: 1600.0
New salary: 41600.0
```

```
Enter current annual salary: 40000
Enter performance rating (1=excellent, 2=good, 3=poor): 3
Raise amount: 600.0
New salary: 40600.0
```

Q38.

```
import java.util.*;
class Q38 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter attendance percentage: ");
        double attendance = input.nextDouble();

        System.out.print("Enter average marks: ");
        double marks = input.nextDouble();

        if (attendance > 80 && marks > 50) {
            System.out.println("You are eligible to sit for the O/L examination.");
        } else {
            System.out.println("You are not eligible to sit for the O/L examination.");
        }
    }
}
```

Outputs:

```
Enter attendance percentage: 81
Enter average marks: 75
You are eligible to sit for the O/L examination.
```

```
Enter attendance percentage: 75
Enter average marks: 50
You are not eligible to sit for the O/L examination.
```

Q39.

```
import java.util.*;
class Q39 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter time : ");
        double time = input.nextDouble();

        if (time < 0.00 || time > 24.00 || (time % 1) > 0.59) {
            System.out.println("Invalid time entered! Minutes cannot exceed 59.");
        } else if (time >= 00.00 && time <= 12.00) {
            System.out.println("Good morning");
        } else if (time > 12.00 && time <= 16.00) {
            System.out.println("Good afternoon");
        } else if (time > 16.00 && time <= 19.00) {
            System.out.println("Good evening");
        } else if (time > 19.00 && time <= 24.00) {
            System.out.println("Good night");
        }
    }
}
```

Outputs:

```
Enter time : 5.00
Good morning
```

```
Enter time : 13.00
Good afternoon
```

```
Enter time : 16.01
Good evening
```

```
Enter time : 19.02
Good night
```

Q40.

```
Q23.java Q24.java Q25.java Q26.java Q27.java Q28.java Q29.java Q30.java Q31.java Q32.java Q33.java Q34.java Q35
import java.util.*;
class Q40 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter year : ");
        int year = input.nextInt();

        System.out.print("Enter month : ");
        String month = input.next();

        boolean is_LeapYear = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);

        if (is_LeapYear) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is not a Leap Year.");
        }

        if (month.equals("January") || month.equals("March") || month.equals("May") || month.equals("July") || month.equals("August") || month.equals("October") || month.equals("December") ||
            month.equals("jannuary") || month.equals("march") || month.equals("may") || month.equals("july") || month.equals("august") || month.equals("october") || month.equals("december")) {
            System.out.println("Number of days in " + month + " " + year + " is: 31");
        }
        else if (month.equals("April") || month.equals("June") || month.equals("September") || month.equals("November") ||
            month.equals("april") || month.equals("june") || month.equals("september") || month.equals("november")) {
            System.out.println("Number of days in " + month + " " + year + " is: 30");
        }
        else if (month.equals("February") ||
            month.equals("february")) {
            if (is_LeapYear) {
                System.out.println("Number of days in February " + year + " is: 29");
            } else {
                System.out.println("Number of days in February " + year + " is: 28");
            }
        }
        else {
            System.out.println("Invalid month name!");
        }
    }
}
```

Outputs:

```
Enter year : 2008
Enter month : january
2008 is a Leap Year.
Number of days in january 2008 is: 31
```

Q41.

```
import java.util.*;
class Q41 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("How many prints do you want : ");
        int p_quan = input.nextInt();

        double price_1 = 30.00;
        double price_2 = 28.00;
        double price_3 = 27.00;
        double price_4 = 26.00;
        double over_price = 25.00;

        if (p_quan >= 0 && p_quan <= 99) {
            System.out.println("Price per copy : Rs." + price_1);
            System.out.println("Total Price : " + (p_quan * price_1));
        }
        else if (p_quan >= 100 && p_quan <= 499) {
            System.out.println("Price per copy : Rs." + price_2);
            System.out.println("Total Price : " + (p_quan * price_2));
        }
        else if (p_quan >= 500 && p_quan <= 799) {
            System.out.println("Price per copy : Rs." + price_3);
            System.out.println("Total Price : " + (p_quan * price_3));
        }
        else if (p_quan >= 800 && p_quan <= 1000) {
            System.out.println("Price per copy : Rs." + price_4);
            System.out.println("Total Price : " + (p_quan * price_4));
        }
        else {
            System.out.println("Price per copy : Rs." + over_price);
            System.out.println("Total Price : " + (p_quan * over_price));
        }
    }
}
```

Outputs:

```
How many prints do you want : 5
Price per copy : Rs.30.0
Total Price : 150.0
```

```
How many prints do you want : 101
Price per copy : Rs.28.0
Total Price : 2828.0
```

```
How many prints do you want : 881
Price per copy : Rs.26.0
Total Price : 22906.0
```

```
How many prints do you want : 501
Price per copy : Rs.27.0
Total Price : 13527.0
```

```
How many prints do you want : 1003
Price per copy : Rs.25.0
Total Price : 25075.0
```

Q42.

```
import java.util.*;
class Q42 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your waist size in inches: ");
        int size = input.nextInt();

        if (size >= 28 && size <= 29) {
            System.out.println("Your size is: X-small (XS)");
        }
        else if (size >= 30 && size <= 31) {
            System.out.println("Your size is: Small (S)");
        }
        else if (size >= 32 && size <= 34) {
            System.out.println("Your size is: Medium (M)");
        }
        else if (size >= 36 && size <= 38) {
            System.out.println("Your size is: Large (L)");
        }
        else if (size >= 40 && size <= 42) {
            System.out.println("Your size is: X-Large (XL)");
        }
        else {
            System.out.println("Size not available in the chart!");
        }
    }
}
```

Outputs:

```
Enter your waist size in inches: 28
Your size is: X-small (XS)
```

```
Enter your waist size in inches: 31
Your size is: Small (S)
```

```
Enter your waist size in inches: 33
Your size is: Medium (M)
```

```
Enter your waist size in inches: 37
Your size is: Large (L)
```

```
Enter your waist size in inches: 41
Your size is: X-Large (XL)
```

Q43.

```
import java.util.*;
class Q43 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the first number      : ");
        int num1 = input.nextInt();

        System.out.print("Enter the second number     : ");
        int num2 = input.nextInt();

        System.out.print("Enter the operator (+,-,*,/,%,^) : ");
        char operator = input.next().charAt(0);

        if (operator == '+') {
            System.out.println(num1 + " + " + num2 + " = " + (num1 + num2));
        }
        else if (operator == '-') {
            System.out.println(num1 + " - " + num2 + " = " + (num1 - num2));
        }
        else if (operator == '*') {
            System.out.println(num1 + " * " + num2 + " = " + (num1 * num2));
        }
        else if (operator == '/') {
            System.out.println(num1 + " / " + num2 + " = " + (num1 / num2));
        }
        else if (operator == '%') {
            System.out.println(num1 + " % " + num2 + " = " + (num1 % num2));
        }
        else if (operator == '^') {
            int result = 1;
            for (int i = 1; i <= num2; i++) {
                result = result * num1;
            }
            System.out.println(num1 + " ^ " + num2 + " = " + result);
        }
        else {
            System.out.println("Invalid operator!");
        }
    }
}
```

Output:

```
Enter the first number      : 5
Enter the second number     : 2
Enter the operator (+,-,*,/,%,^) : ^
5 ^ 2 = 25
```

Q44.

```
import java.util.*;
class Q44 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a number from 1 to 28 : ");
        int day = input.nextInt();

        if (day == 1 || day == 8 || day == 15 || day == 22) {
            System.out.println("Tuesday");
        }
        else if (day == 2 || day == 9 || day == 16 || day == 23) {
            System.out.println("Wednesday");
        }
        else if (day == 3 || day == 10 || day == 17 || day == 24) {
            System.out.println("Thursday");
        }
        else if (day == 4 || day == 11 || day == 18 || day == 25) {
            System.out.println("Friday");
        }
        else if (day == 5 || day == 12 || day == 19 || day == 26) {
            System.out.println("Saturday");
        }
        else if (day == 6 || day == 13 || day == 20 || day == 27) {
            System.out.println("Sunday");
        }
        else if (day == 7 || day == 14 || day == 21 || day == 28) {
            System.out.println("Monday");
        }
        else {
            System.out.println("Invalid number! Please enter between 1 and 28.");
        }
    }
}
```

Output:

```
Enter a number from 1 to 28 : 5
Saturday
```

Q45.

```
import java.util.Scanner;

class Q45 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter row number (1-8): ");
        int row = input.nextInt();

        System.out.print("Enter column number (1-8): ");
        int col = input.nextInt();

        if (row >= 1 && row <= 8 && col >= 1 && col <= 8) {
            if ((row % 2 == 0 && col % 2 == 0) || (row % 2 != 0 && col % 2 != 0)) {
                System.out.println("The color of the square is: White");
            } else {
                System.out.println("The color of the square is: Black");
            }
        } else {
            System.out.println("Invalid input! Row and column must be between 1 and 8.");
        }
    }
}
```

Output:

```
Enter row number (1-8): 6
Enter column number (1-8): 7
The color of the square is: Black
```

```
Enter row number (1-8): 4
Enter column number (1-8): 6
The color of the square is: White
```

Q46.

```
import java.util.*;
class Q46 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter first number (a): ");
        int a = input.nextInt();

        System.out.print("Enter second number (b): ");
        int b = input.nextInt();

        System.out.print("Enter third number (c): ");
        int c = input.nextInt();

        if ((a * a) + (b * b) == (c * c) || (a * a) + (c * c) == (b * b) || (b * b) + (c * c) == (a * a)) {
            System.out.println("They form a Pythagorean triple.");
        } else {
            System.out.println("They do not form a Pythagorean triple.");
        }
    }
}
```

Output:

```
Enter first number (a): 3
Enter second number (b): 4
Enter third number (c): 5
They form a Pythagorean triple.
```

Q47.

```
import java.util.*;
class Q47 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter month (1-12): ");
        int month = input.nextInt();

        System.out.print("Enter day (1-31): ");
        int day = input.nextInt();

        if ((month == 12 && day >= 21) || month == 1 || month == 2 || (month == 3 && day <= 19)) {
            System.out.println("Season: Winter");
        }
        else if ((month == 3 && day >= 20) || month == 4 || month == 5 || (month == 6 && day <= 20)) {
            System.out.println("Season: Spring");
        }
        else if ((month == 6 && day >= 21) || month == 7 || month == 8 || (month == 9 && day <= 21)) {
            System.out.println("Season: Summer");
        }
        else if ((month == 9 && day >= 22) || month == 10 || month == 11 || (month == 12 && day <= 20)) {
            System.out.println("Season: Autumn");
        }
        else {
            System.out.println("Invalid date or month entered!");
        }
    }
}
```

Output:

```
Enter month (1-12): 3
Enter day (1-31): 4
Season: Winter
```

Q48.

```
import java.util.*;
class Q48 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter month of birth (1-12): ");
        int month = input.nextInt();

        System.out.print("Enter day of birth (1-31): ");
        int day = input.nextInt();

        if ((month == 12 && day >= 22) || (month == 1 && day <= 19)) {
            System.out.println("Astrological sign: Capricornus");
        }
        else if ((month == 1 && day >= 20) || (month == 2 && day <= 18)) {
            System.out.println("Astrological sign: Aquarius");
        }
        else if ((month == 2 && day >= 19) || (month == 3 && day <= 20)) {
            System.out.println("Astrological sign: Pisces");
        }
        else if ((month == 3 && day >= 21) || (month == 4 && day <= 19)) {
            System.out.println("Astrological sign: Aries (Ram)");
        }
        else if ((month == 4 && day >= 20) || (month == 5 && day <= 20)) {
            System.out.println("Astrological sign: Taurus");
        }
        else if ((month == 5 && day >= 21) || (month == 6 && day <= 21)) {
            System.out.println("Astrological sign: Gemini");
        }
        else if ((month == 6 && day >= 22) || (month == 7 && day <= 22)) {
            System.out.println("Astrological sign: Cancer");
        }
        else if ((month == 7 && day >= 23) || (month == 8 && day <= 22)) {
            System.out.println("Astrological sign: Leo");
        }
        else if ((month == 8 && day >= 23) || (month == 9 && day <= 22)) {
            System.out.println("Astrological sign: Virgo");
        }
        else if ((month == 9 && day >= 23) || (month == 10 && day <= 23)) {
            System.out.println("Astrological sign: Libra");
        }
        else if ((month == 10 && day >= 24) || (month == 11 && day <= 21)) {
            System.out.println("Astrological sign: Scorpius");
        }
        else if ((month == 11 && day >= 22) || (month == 12 && day <= 21)) {
            System.out.println("Astrological sign: Sagittarius");
        }
        else {
            System.out.println("Invalid date!");
        }
    }
}
```

Output:

```
Enter month of birth (1-12): 3
Enter day of birth (1-31): 6
Astrological sign: Pisces
```

```
-----
(program exited with code: 0)
```

Q49.

```

import java.util.*;
class Q49 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter basic salary: ");
        double basic = input.nextDouble();

        double housing_pe = 0;
        double travel_pe = 0;

        if (basic <= 10000) {
            housing_pe = 0.20;
            travel_pe = 0.60;
        }
        else if (basic <= 20000) {
            housing_pe = 0.25;
            travel_pe = 0.70;
        }
        else {
            housing_pe = 0.30;
            travel_pe = 0.75;
        }

        double h_allowance = basic * housing_pe;
        double t_allowance = basic * travel_pe;
        double gross_Salary = basic + h_allowance+ t_allowance;

        System.out.println("Housing Allowance (HA): " + h_allowance);
        System.out.println("Travel Allowance (TA): " + t_allowance);
        System.out.println("Gross Salary: " + gross_Salary);

    }
}

```

Output:

```

Enter basic salary: 40000
Housing Allowance (HA): 12000.0
Travel Allowance (TA): 30000.0
Gross Salary: 82000.0

```

Q50.

```
import java.util.*;
class Q50 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter angle a: ");
        int a = input.nextInt();

        System.out.print("Enter angle b: ");
        int b = input.nextInt();

        System.out.print("Enter angle c: ");
        int c = input.nextInt();

        if (a > 0 && b > 0 && c > 0 && (a + b + c == 180)) {
            System.out.println("The triangle can be formed.");
        } else {
            System.out.println("The triangle cannot be formed.");
        }
    }
}
```

Outputs:

```
Enter angle a: 45
Enter angle b: 65
Enter angle c: 70
The triangle can be formed.
```

```
Enter angle a: 34
Enter angle b: 56
Enter angle c: 24
The triangle cannot be formed.
```

Q51.

```
import java.util.*;
class Q51 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = input.nextInt();

        if (age > 65) {
            System.out.println("Classification: Senior");
        }
        else if (age > 20 && age <= 65) {
            System.out.println("Classification: Adult");
        }
        else if (age > 13 && age <= 20) {
            System.out.println("Classification: Teenager");
        }
        else if (age > 1 && age <= 13) {
            System.out.println("Classification: Child");
        }
        else if (age <= 1 && age >= 0) {
            System.out.println("Classification: Infant");
        }
        else {
            System.out.println("Invalid age entered!");
        }
    }
}
```

Output:

```
Enter your age: 65
Classification: Adult
```

```
Enter your age: 1
Classification: Infant
```

Q52.

```
import java.util.*;
class Q52 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter x coordinate: ");
        double x = input.nextDouble();

        System.out.print("Enter y coordinate: ");
        double y = input.nextDouble();

        if (x > 0 && y > 0) {
            System.out.println("The point lies in Quadrant I");
        }
        else if (x < 0 && y > 0) {
            System.out.println("The point lies in Quadrant II");
        }
        else if (x < 0 && y < 0) {
            System.out.println("The point lies in Quadrant III");
        }
        else if (x > 0 && y < 0) {
            System.out.println("The point lies in Quadrant IV");
        }
        else if (x == 0 && y == 0) {
            System.out.println("The point lies at the Origin");
        }
        else if (x == 0) {
            System.out.println("The point lies on the Y-axis");
        }
        else {
            System.out.println("The point lies on the X-axis");
        }
    }
}
```

Outputs:

```
Enter x coordinate: 8
Enter y coordinate: 0
The point lies on the X-axis
```

```
Enter x coordinate: 0
Enter y coordinate: 6
The point lies on the Y-axis
```

```
Enter x coordinate: 0
Enter y coordinate: 0
The point lies at the Origin
```

```
Enter x coordinate: 4
Enter y coordinate: -7
The point lies in Quadrant IV
```

```
Enter x coordinate: -3
Enter y coordinate: 4
The point lies in Quadrant II
```

```
Enter x coordinate: -3
Enter y coordinate: -5
The point lies in Quadrant III
```

```
Enter x coordinate: 5
Enter y coordinate: 2
The point lies in Quadrant I
```

Q53.

```
import java.util.*;
class Q53 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter Systolic Blood Pressure (SBP): ");
        int sbp = input.nextInt();

        System.out.print("Enter Diastolic Blood Pressure (DBP): ");
        int dbp = input.nextInt();

        if (sbp >= 130 && dbp >= 90) {
            System.out.println("Status: High Pressure");
        }
        else if (sbp <= 100 && dbp <= 70) {
            System.out.println("Status: Low Pressure");
        }
        else if (sbp >= 100 && sbp <= 130 && dbp >= 70 && dbp <= 90) {
            System.out.println("Status: Normal");
        }
    }
}
```

Output:

```
Enter Systolic Blood Pressure (SBP): 90
Enter Diastolic Blood Pressure (DBP): 60
Status: Low Pressure
```

```
Enter Systolic Blood Pressure (SBP): 150
Enter Diastolic Blood Pressure (DBP): 100
Status: High Pressure
```

```
Enter Systolic Blood Pressure (SBP): 101
Enter Diastolic Blood Pressure (DBP): 80
Status: Normal
```

Q54.

```
import java.util.*;
class Q54 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the URL: ");
        String url = input.next();

        if (url.endsWith(".com")) {
            System.out.println("Website type: Commercial website");
        }
        else if (url.endsWith(".org")) {
            System.out.println("Website type: Organization website");
        }
        else if (url.endsWith(".lk")) {
            System.out.println("Website type: Sri Lankan website");
        }
        else {
            System.out.println("Unknown website type!");
        }
    }
}
```

Outputs:

```
Enter the URL: .org
Website type: Organization website
```

```
1 Enter the URL: .com
Website type: Commercial website
```

```
3 Enter the URL: .lk
Website type: Sri Lankan website
```

Q55.

```
import java.util.*;
class Q55 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter the number to check: ");
        double number = input.nextDouble();

        System.out.print("Enter the lower bound: ");
        double lowerBound = input.nextDouble();

        System.out.print("Enter the upper bound: ");
        double upperBound = input.nextDouble();

        if (number >= lowerBound && number <= upperBound) {
            System.out.println("The number is within the given range.");
        } else {
            System.out.println("The number is outside the given range.");
        }
    }
}
```

Outputs:

```
Enter the number to check: 45
Enter the lower bound: 34
Enter the upper bound: 199
The number is within the given range.
```

```
Enter the number to check: 45
Enter the lower bound: 46
Enter the upper bound: 399
The number is outside the given range.
```

Q56.

Answer: $item = (a \geq 10) \ \&\& \ (b < 50);$

Q57.

```
import java.util.*;
class Q57 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int month_1 = 0, day_1 = 0, year_1 = 0;
        int month_2 = 0, day_2 = 0, year_2 = 0;

        System.out.println("Enter Date 1:");
        System.out.print("Month (1-12): ");
        month_1 = input.nextInt();
        System.out.print("Day (1-31): ");
        day_1 = input.nextInt();
        System.out.print("Year: ");
        year_1 = input.nextInt();

        System.out.println("\nEnter Date 2:");
        System.out.print("Month (1-12): ");
        month_2 = input.nextInt();
        System.out.print("Day (1-31): ");
        day_2 = input.nextInt();
        System.out.print("Year: ");
        year_2 = input.nextInt();

        if (year_1 < year_2) {
            System.out.println("\nDate 1 comes first.");
        }
        if (year_2 < year_1) {
            System.out.println("\nDate 2 comes first.");
        }
        if (year_1 == year_2) {
            if (month_1 < month_2) {
                System.out.println("\nDate 1 comes first.");
            }
            if (month_2 < month_1) {
                System.out.println("\nDate 2 comes first.");
            }
            if (month_1 == month_2) {
                if (day_1 < day_2) {
                    System.out.println("\nDate 1 comes first.");
                }
                if (day_2 < day_1) {
                    System.out.println("\nDate 2 comes first.");
                }
                if (day_1 == day_2) {
                    System.out.println("\nBoth dates are the same.");
                }
            }
        }
    }
}
```

Output:

```
Enter Date 1:
Month (1-12): 4
Day (1-31): 6
Year: 2026

Enter Date 2:
Month (1-12): 5
Day (1-31): 7
Year: 2025

Date 2 comes first.
```

Q58.

```
import java.util.*;
class Q58 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter year: ");
        int y = input.nextInt();

        System.out.print("Enter month: ");
        int m = input.nextInt();

        System.out.print("Enter day: ");
        int d = input.nextInt();

        int y_cal = y - ((14 - m) / 12);
        int x = y_cal + y_cal / 4 - y_cal / 100 + y_cal / 400;
        int m_cal = m + 12 * ((14 - m) / 12) - 2;
        int d_cal = (d + x + (31 * m_cal) / 12) % 7;

        if (d_cal == 0) {
            System.out.println("Day of the week: Sunday");
        }
        if (d_cal == 1) {
            System.out.println("Day of the week: Monday");
        }
        if (d_cal == 2) {
            System.out.println("Day of the week: Tuesday");
        }
        if (d_cal == 3) {
            System.out.println("Day of the week: Wednesday");
        }
        if (d_cal == 4) {
            System.out.println("Day of the week: Thursday");
        }
        if (d_cal == 5) {
            System.out.println("Day of the week: Friday");
        }
        if (d_cal == 6) {
            System.out.println("Day of the week: Saturday");
        }
    }
}
```

Output:

```
Enter year: 2006
Enter month: 12
Enter day: 4
Day of the week: Monday
```

Q59.

```
import java.util.*;
class Q59 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter R values (0 to 255): ");
        int r = input.nextInt();
        System.out.print("Enter G values (0 to 255): ");
        int g = input.nextInt();
        System.out.print("Enter B values (0 to 255): ");
        int b = input.nextInt();

        double c, m, y, k;

        if (r == 0 && g == 0 && b == 0) {
            c = 0.0;
            m = 0.0;
            y = 0.0;
            k = 1.0;
        } else {
            double rPrime = r / 255.0;
            double gPrime = g / 255.0;
            double bPrime = b / 255.0;

            double w = Math.max(rPrime, Math.max(gPrime, bPrime));

            c = (w - rPrime) / w;
            m = (w - gPrime) / w;
            y = (w - bPrime) / w;
            k = 1 - w;
        }
        System.out.println("Cyan = " + c);
        System.out.println("Magenta = " + m);
        System.out.println("Yellow = " + y);
        System.out.println("black = " + k);
    }
}
```

Output:

```
Enter R values (0 to 255): 255
Enter G values (0 to 255): 123
Enter B values (0 to 255): 0
Cyan = 0.00000
Magenta =0.51765
Yellow = 1.00000
black = 0.00000
```

Q60.

```
import java.util.*;
public class Q60 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter parcel weight in kg: ");
        double weight_val = input.nextDouble();

        double totalCharge = 0;

        if (weight_val <= 5) {
            totalCharge = 500;
        } else {
            double ex_Weight = weight_val - 5;
            totalCharge = 500 + (ex_Weight * 100);
        }

        System.out.println("Total courier charge: Rs. " + totalCharge);
    }
}
```

Output:

```
Enter parcel weight in kg: 67
Total courier charge: Rs. 6700.0
```